# INTERNSHIP PROJECT REPORT

**Project Title:** Password Strength Analyzer with Custom Wordlist Generator(PassForge)

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**Tools Used:** Python, VS Code, GitHub

## Introduction

Passwords are a primary layer of defense in digital systems. Weak or reused passwords make systems vulnerable to cyberattacks such as brute force or credential stuffing. Hence, strong password practices and early breach detection are essential.PassForge is a command-line cybersecurity tool designed to help users:

- Analyze password strength
- Detect if a password has been breached
- Generate custom wordlists for penetration testing

## Abstract

This project simulates real-world security tool development using Python. PassForge enables users to input passwords and receive feedback on strength, crack time, and whether their password appears in known data leaks (using the HaveIBeenPwned API). The second functionality helps ethical hackers by generating strong, mutated password wordlists using keywords provided by the user.

Through this project, I learned practical Python scripting, API integration, command-line interface design, and modular coding — all essential for building effective security tools.

## Tools & Libraries Used

| | | |
|---|---|---|
| Python 3.x | - | Core programming language |
| Argparse | - | Handle command-line arguments |
| Zxcvbn | - | Analyze password strength |
| Requests | - | API integration (breach check) |
| Pyfiglet | - | ASCII art banners |
| GitHub | - | Version control & hosting |

VS Code - Development environment

## Steps Involved in Building the Project

1. **Planning Functionality**
   - Defined two core functions: password analysis & wordlist generation
   - Designed simple CLI using flags: `--analyze`, `--generate-wordlist`
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2. **Code Structure**
   - Created **`main.py`** as CLI entry point
   - Made separate modules:
     - `analyzer.py`: - Password scoring, suggestions
     - `breach_checker.py`: - HIBP API hash check
     - `generator.py`: - Wordlist with mutations
3. **Password Strength Analysis**
   - Used `zxcvbn` library to rate strength and suggest improvements
   - Displayed estimated crack time (online/offline)
4. **Breach Detection**
   - Converted password to SHA1 and used only prefix to maintain privacy
   - Called HaveIBeenPwned API to detect breach status
5. **Wordlist Generation**
   - Took keywords from user input
   - Applied mutation techniques:
     - Leetspeak (e.g., a→@, i→1)
     - Capitalization, reversals, year suffixes
   - Saved results in `wordlist.txt`
6. **Testing & Finalization**
   - Ran test cases with common and strong passwords
   - Improved CLI outputs with banners and messages
   - Final code cleanup and `.gitignore` for clean repo

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## Conclusion

PassForge is a practical cyber security project that helped me gain hands-on experience with real-world security concerns. It taught me how to:

- Build modular Python tools
- Work with APIs securely
- Handle CLI operations like professionals

The tool can be used in ethical hacking labs, CTFs, and as a learning reference for password security. This internship project enhanced both my technical skills and understanding of secure coding practices.

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